READ ME

Research Data Management for Agrosystem Sciences -

Modular Training Material for Reuse

Authors

Sophie Boße (ORCID: 0009-0002-6461-8291)[1a], Elena Rey Mazón (ORCID: 0000-0003-4813-5927)[2b], Wahib Sahwan (ORCID: 0000-0002-6503-5525)[3c], Lea Sophie Singson (ORCID: 0009-0004-9978-8703)[4d], Lucia Vedder (ORCID: 0000-0002-8924-9800)[5e]

[1]ZB MED - Information Centre for Life Sciences

[2]Leibniz-Institut für Pflanzengenetik und Kulturpflanzenforschung (IPK) Gatersleben

[3]Leibniz-Zentrum für Agrarlandschaftsforschung (ZALF) e.V.

[4]Leibniz-Zentrum für Informationsinfrastruktur (FIZ) Karlsruhe

[5]Rheinische Freidrich-Wilhelms-Universität Bonn

[a]Overall concept and content responsibility

[b]Contents responsibility for the following topics: 00_Welcome, 01_Researchdatamanagement, 04_FAIR-Principles, 08_DataCollection

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Version

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License

Discipline

Agrosystem Science Research

Language

English

Description

This is an extensive Open Educational Resource (OER) comprising training materials designed to help you plan, create and conduct training events on Research Data Management (RDM) for agrosystem science researchers. These resources include presentation slides, presenter notes, teaching scripts with learning objectives, worksheets and a recording of an RDM training session, as well as a guide on how to create a training session. All resources are structured and described in accordance with the EduBrick concept, whereby the teaching resources are broken down into the smallest didactic unit, called 'Bricks', which can be used to create targeted learning/teaching units for training events tailored to the respective needs. To enhance reusability, all materials are provided in multiple formats and are openly licensed.

We also provide an example of how the Bricks could be used to create a four-hour online training course, including a teaching script, presentation slides, worksheets and a video recording of the training course.

Keywords

Research Data Management, Open Educational Resource, EduBricks, Agrosystem Science Research, FAIRagro, Training material

Target Group

The target group of this resource are RDM professionals, data stewards, and teachers in agrosystem science, who want to conduct discipline-specific RDM training for researchers within the agrosystem sciences.

Intended Use

This publication is designed to:

- Enable flexible reuse and recombination
- Support trainers in different institutional contexts
- Facilitate adaptation to specific target groups
- Encourage collaborative use without structural modification

The modular design ensures that content can be tailored to specific formats, disciplines, and levels of expertise while maintaining didactical coherence.

Content

This publication includes:

- READ ME file (pdf)
- "How to create a training" Guide (pdf)
- Recording of an four hour example training session (mp4)
- Three Zip folders containing all training materials (presentation slides, teaching scripts, worksheets) in multiple formats:
    - "Bricks_pdf": all materials in pdf formats (excluding presenters notes)
    - "Bricks_MicrosoftOffice": presentation slides as pptx, teachingscript as xlsx, worksheets as docx
    - "Bricks_LibreOffice": presentation slides as odp, teaching script as ods, worksheets as odt

We recommend using Microsoft Office or LibreOffice, depending on your system, rather than using PDF files.

Slide Design and Visual Structure

To ensure clarity and usability across contexts, the slides follow a consistent visual logic:

Color Coding

There is a coloured mark in the top right corner of every slide:

- Petrol – Generic content (discipline-independent RDM topics)
- Light green – Agrosystem-specific content
- Red-orange – Interactive content (exercises, discussions, reflection prompts)

Logo Usage

The FAIRagro logo appears only:

- On title slides
- On section divider slides

This deliberate limitation facilitates reuse in collaborative contexts, as cooperating institutions do not need to include their logos into every slide.

Concept of Modularity

The training material is designed as a modular system. It is inspired by the concept of “EduBricks” of the National Research Data Infrastructure Section Education & Training (Brilhaus et al., 2025).

Bricks – The Smallest Didactic Unit

All content is broken down into individual “bricks”, which represent the smallest didactic unit. A brick may contain:

- Generic content
- Discipline-specific content
- Interactive elements

Each brick:

- Has its own learning objective(s), which are taken from or adapted from the learning objectives matrix for research data management (Petersen et al., 2025)
- Has a unique ID
- Consists of presentation slides with detailed speaker notes
- May include additional materials (e.g. worksheets)

Bricks are stored in topic-based folders.

Many bricks can be used in both online and in-person formats.
Where necessary, format-specific variants are provided and clearly marked.

From Bricks to Units to Disseminations

The modular structure enables flexible recombination:

- Bricks → combined into thematic Units (via didactic learning paths)
- Units → combined into a Dissemination (i.e. a full training session)

The composition depends on:

- Target group
- Training format
- Available time
- Learning objectives

Folder Structure

The folder structure within the Zip folders reflects the thematic organization and numbering logic.

For the best display, we recommend sorting by name, regardless of file type.

The following topics and subtopics are included in the folder structure:

00_Welcome
  00_Welcome_Teachingscript
  00-01_Titel
  00-02_FAIRagro
  00-03_AboutThisCourse
  00-04_Agenda
  00-05_Icebreaker
  00-06_ResearchJourney_Slides
01_Researchdatamangement
  01_Researchdatamanagement_Teachingscript
  01-01_Researchdata
    01-01-06_FileFormats
  01-02_Researchdatamanagement
  01-03_DataLifecycle
02_GoodResearchPractice
03_ResearchPolicies
04_FAIR-Prinzipien
05_DataOrganisation
  05-02_DataOrganisation_NamingConvention
  05-03_DataOrganisation_FolderStructure
  05-04_DataOrganisation_VersionControl

06_DataDocumentation
- 06-01_DataDocumentation_Intro
- 06-02_DataDocumentation_Types
- 06-03_DataDocumentation_Metadaten
- 06-04_DataDocumentation_Metadatenstandards
- 06-05_DataDocumentation_SemanticWeb
- 06-06_DataDocumentation_PersistentIdentifier
- 06-07_DataDocumentation_ComputationalReproducibility

07_DataManagementPlan
- 07-07_DataManagementPlan_Tools

08_DataCollection
- 08-01_DataCollection_ElectronicLabNotebooks
- 08-02_DataCollection_ElectronicFieldbooks
- 08-03_DataCollection_DataReuse

09_DataStorage

10_DataPublication
- 10-01_DataPublication_Sharing
- 10-02_DataPublication_Repositories
- 10-03_DataPublication_Data-Journals

11_LegalAspects
- 11-00_LegalAspects_CaseStudy
- 11-01_LegalAspects_Datarights
- 11-02_LegalAspects_Dataprotection
- 11-03_LegalAspects_BusinessSecrets
- 11-04_LegalAspects_Nagoya
- 11-05_LegalAspects_Structure

12_WrapUp

Template_Teachingscript

FAIRagro_Example_Training

HowToCreateATraining

README

ID Nomenclature

Each brick follows a consistent ID structure:

XX-XX-XX-XXx

This corresponds to the numbers of:

TOPIC_SUBTOPIC_SUBSUBTOPIC_BRICKvariant

- The number of subtopics varies.
- The numerical structure reflects the thematic hierarchy.
- Variants (e.g., online vs. in-person) are marked with a lowercase letter (e.g., *a*, *b*).
- The ID is included in the file name.

- The ID links directly to the detailed description in the corresponding Teachingscript.

The Teaching Script

Each main topic folder contains a teaching script.

For every brick, the teaching script provides:

- ID
- Topic
- Subtopic
- Learning objective(s)
  - Adapted from the learning objectives matrix for research data management (Petersen et al. 2025)
- Content description
- Intended duration
- Suggested timeline (including the time by which the topic should be completed)
- Format (Form of content delivery e.g. Brainstorming, Discussion, Exercise, Presentation)
- Required materials
- Additional comments
- Didactic concept ("breathing in and breathing out")

Didactic Concept: Breathing In and Breathing Out

The pedagogical approach alternates between:

- Breathing in – Phases of information input
- Breathing out – Phases of reflection, discussion, or active engagement

A well-designed training alternates these phases to maintain attention and support knowledge consolidation. (Biernacka et al.)

How to Assemble a Training Session

The file "HowToCreateATraining" provides a step-by-step guide on how to:

- Define the training format
- Create the content structure
- Select appropriate bricks
- Adapt content to format and audience
- Design and deliver a coherent training session

Example Training

An example of a four hour training format, based on this modular training material, is included. For this example the following materials are included:

- Teaching script
- Presentation slides
- Worksheets
- Link collection to share with participants
- Announcement text
- Recording of the non-interactive parts of a training delivered by FAIRagro Data Stewards

References

Biernacka, Katarzyna, et al. “Train-the-Trainer-Konzept zum Thema Forschungsdatenmanagement.” Zenodo, 14 Dec. 2023. *DOI.org (Datacite)*, https://doi.org/10.5281/ZENODO.1215376.

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Petersen, Britta, et al. *Learning Objectives Matrix on the Topic of Research Data Management (RDM)*. July 2025. *DOI.org (Datacite)*, https://doi.org/10.5281/ZENODO.15846806.